# PES UNIVERSITY

Department of Computer Science & Engineering

LAB 1: REQUIREMENTS ENGINEERING & UML USE-CASE MODELLING

Problem Statement #29 – Municipal Infrastructure Damage Reporting App

| | |
|---|---|
| **Name** | Dinesh |
| **SRN** | PES1UG24CS157 |
| **Subject** | Software Engineering (UE24CS341A) |
| **Semester** | Sem-5 |
| **Section** | C |

Academic Year: 2026–27

# 1. Problem Context & Overview

The Municipal Infrastructure Damage Reporting App is a citizen portal for reporting road potholes, broken streetlights, and water pipeline leaks using geotagged photos. The system automatically dispatches submitted tickets to the relevant municipal engineering departments.
Target stakeholders/actors identified in the assigned problem statement: Citizen Reporter and Municipal Engineer.

# 2. Requirements Table

| Req ID | Type | Description | Priority | Acceptance Criteria | Rationale |
|---|---|---|---|---|---|
| FR-001 | Functional | The system shall extract EXIF GPS metadata from uploaded damage photos to automatically pinpoint the incident location on the municipal GIS map. | High | Pass: A photo containing valid EXIF GPS data produces a map location matching the embedded coordinates and the report is routed to the corresponding municipal division. Fail: the system accepts a location-dependent report without location data or flags the location incorrectly. | Provides automatic, accurate geolocation and supports correct departmental routing. |
| FR-002 | Functional | The system shall allow a citizen reporter to submit a damage report containing a damage category, description, and at least one damage photo. | High | Pass: A report is created only after the category, description, and photo are present; a confirmation with a unique ticket ID is displayed after successful submission. Fail: incomplete reports are submitted. | Captures the minimum evidence needed to investigate municipal infrastructure damage. |
| FR-003 | Functional | The system shall automatically dispatch each submitted damage ticket to the relevant municipal engineering department based on the reported damage category and location. | High | Pass: Road pothole, streetlight, and water-pipeline reports are routed to their configured responsible divisions using the category/location rules. Fail: a ticket is left unassigned or routed to an unrelated division. | Reduces manual triage and ensures that incidents reach the responsible engineering team. |
| FR-004 | Functional | The system shall allow a municipal engineer to review an assigned ticket and update its status and resolution notes. | High | Pass: An authorized engineer can change a ticket through the configured workflow (for example, Assigned, In Progress, Resolved) and save a timestamped resolution note. Fail: an unauthorized user can modify the ticket. | Supports operational handling, accountability, and resolution tracking. |
| FR-005 | Functional | The system shall allow a citizen reporter to view the current status of a submitted ticket and receive a submission or status-update notification. | Medium | Pass: Entering a valid ticket ID/account reference displays the latest status, and a notification is generated when the ticket is submitted or its status changes. Fail: the citizen cannot retrieve a valid ticket or receives no status update. | Improves transparency and keeps the reporter informed about municipal action. |
| NFR-001 | Nonfunctional – Performance & Security | The reporting portal shall support offline report drafting and auto-sync when mobile network connectivity resumes. | High | Pass: During an offline test, a citizen can save a draft locally without data loss; when connectivity resumes, the queued report synchronizes automatically and reaches the server within 30 seconds under normal load. Fail: the draft is lost or remains unsynchronized after connectivity resumes. | Ensures reporting remains usable in areas with unreliable mobile connectivity. |
| NFR-002 | Nonfunctional – Performance & Availability | The system shall respond to normal report submission and ticket-status requests within 3 seconds for at least 95% of requests under the defined peak-load test. | High | Pass: A peak-load test shows at least 95% of report/status requests completing within 3 seconds, with no critical service failure. Fail: fewer than 95% meet the 3-second threshold. | Provides responsive service during periods of high citizen usage. |

## 3. Actors

| Actor | Role |
|---|---|
| Citizen Reporter | Reports potholes, broken streetlights, and water pipeline leaks; tracks submitted tickets and may add evidence. |
| Municipal Engineer | Reviews assigned tickets, updates status, and records resolution information. |
| Municipal GIS | Provides map/location services used to pinpoint and route incidents. |
| Notification Service | Delivers submission and ticket-status notifications to citizens. |

## 4. Use Cases

| ID | Use Case | Primary Actor(s) |
|---|---|---|
| UC-01 | Report Damage | Citizen Reporter |
| UC-02 | Upload Damage Photo | Citizen Reporter |
| UC-03 | Extract GPS & Pinpoint Location | Citizen Reporter / Municipal GIS |
| UC-04 | Dispatch Ticket | Municipal GIS / Notification Service |
| UC-05 | Review & Update Ticket | Municipal Engineer |
| UC-06 | Track Report Status | Citizen Reporter / Notification Service |
| UC-07 | Add Additional Evidence | Citizen Reporter |

## 5. UML Use-Case Diagram

UC-01 includes photo upload and GPS/location extraction because these are required parts of the reporting workflow. UC-07 extends UC-01 because adding additional evidence is optional.

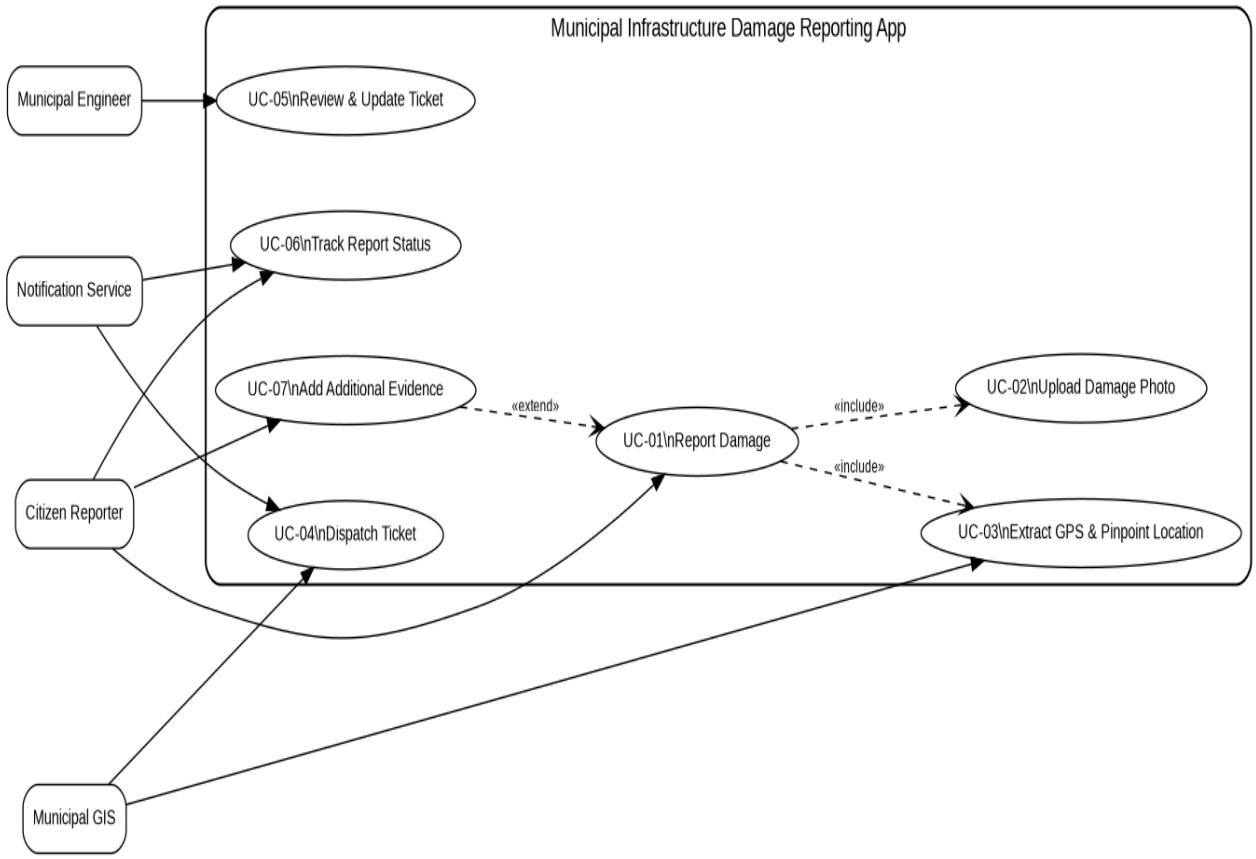



Figure 1: UML Use-Case Diagram

# 6. Use-Case Flow Specification

## UC-01: Report Damage

**Goal:** Enable a citizen to submit a geotagged municipal infrastructure damage report and obtain a ticket for municipal action.
**Primary Actor:** Citizen Reporter
**Supporting Actors:** Municipal GIS, Municipal Engineer, Notification Service
**Preconditions:** The portal is available, or offline drafting is possible; the citizen has damage evidence; routing rules are configured.
**Postconditions:** A valid report has a unique ticket ID, location information is stored when available, the ticket is dispatched, and a notification is issued when connectivity permits.

## Main Success Scenario

1. Citizen Reporter selects the option to report infrastructure damage.
2. System displays the reporting form with damage category, description, photo upload, and location information.
3. Citizen selects the damage category (pothole, broken streetlight, or water pipeline leak) and enters a description.
4. Citizen uploads a damage photo.
5. System extracts EXIF GPS metadata from the photo and pinpoints the incident on the municipal GIS map.
6. System validates the report contents and creates a unique ticket.
7. System automatically dispatches the ticket to the relevant municipal engineering department based on category and location.
8. System confirms submission and provides the ticket ID; the notification service sends a confirmation/status notification.
9. Use case ends successfully.

## Alternate Flow – AF-01: Missing/Invalid GPS Metadata

1. At Step 5, the system detects that the uploaded photo has no valid EXIF GPS metadata.
2. The system informs the citizen that the photo does not contain usable location data and requests a valid location.
3. Citizen supplies an accepted location through the available location mechanism.
4. System validates the location and continues from Step 6 of the Main Success Scenario.
5. If no acceptable location is supplied, the report remains as a draft and is not dispatched until the location requirement is satisfied.

# 7. Requirement-to-Model Traceability

| Requirement | Related Use Case(s) |
|---|---|
| FR-001 | UC-01, UC-03 |
| FR-002 | UC-01, UC-02 |
| FR-003 | UC-01, UC-04 |
| FR-004 | UC-05 |
| FR-005 | UC-06 |
| NFR-001 | UC-01, UC-06 |
| NFR-002 | Applies to all user-facing use cases |

# 8. Conclusion

The requirements and UML model define a complete core workflow for reporting municipal infrastructure damage. The model covers citizen reporting, geotagged evidence, automatic ticket dispatch, municipal engineering action, status tracking, and optional additional evidence. The acceptance criteria are written as measurable pass/fail conditions to support verification during testing.

**Source basis:** PES University, Dept. of CSE, Lab 1 – Requirements Engineering & UML Use-Case Modelling, Problem Statement #29.